

Module 3 - Grad Cafe SQL Query Results

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Database: PostgreSQL table **applicants**, 50,000 rows loaded from Module 2's

llm_extend_applicant_data.json. Queries executed with psql on September 16, 2026.

Formatting: counts are whole numbers; percentages and averages use two decimals. Text matching uses ILIKE / case-insensitive regex. For averages, an applicant 'provides' a metric only when the stored value is a possible score on that scale (GRE Q/V 130-170, GRE AW 0-6, GPA 0-4.0) - see Question 3 for why.

Question 1: Fall 2026 applicant count

Question. How many entries in the database are from applicants who applied for Fall 2026?

Result. Fall 2026 applicant count: 33,207

SQL.

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE 'fall 2026';
```

Explanation. Counts every row whose term column reads 'Fall 2026' (case-insensitive ILIKE).

Question 2: Percent international

Question. Among entries that provide a nationality classification, what percentage are international students?

Result. Percent international: 47.89%

SQL.

```
SELECT ROUND(100.0 * COUNT(*) FILTER (WHERE us_or_international ILIKE 'international')
            / NULLIF(COUNT(*), 0), 2)
FROM applicants
WHERE us_or_international IS NOT NULL AND BTRIM(us_or_international) <> '';
```

Explanation. The WHERE clause restricts the denominator to rows with a usable nationality value (NULL/blank excluded). The FILTER clause counts only 'International' rows for the numerator; 'American' and 'Other' therefore fall in the denominator but not the numerator. NULLIF guards against division by zero.

Question 3: Average GPA / GRE / GRE V / GRE AW

Question. What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result. Average GPA: 3.75

Average GRE Quantitative: 165.75

Average GRE Verbal: 160.37

Average GRE Analytical Writing: 4.33

SQL.

```
SELECT ROUND(AVG(gpa) FILTER (WHERE gpa > 0 AND gpa <= 4.0)::numeric, 2) AS avg_gpa,
       ROUND(AVG(gre) FILTER (WHERE gre BETWEEN 130 AND 170)::numeric, 2) AS avg_gre,
       ROUND(AVG(gre_v) FILTER (WHERE gre_v BETWEEN 130 AND 170)::numeric, 2) AS avg_gre_v,
       ROUND(AVG(gre_aw) FILTER (WHERE gre_aw BETWEEN 0 AND 6)::numeric, 2) AS avg_gre_aw
FROM applicants;
```

Explanation. Each average is computed separately over only the applicants who supplied that metric: AVG() ignores NULLs, and the FILTER clause additionally requires the value to be a possible score on that scale (GRE Q/V 130-170, GRE AW 0-6, GPA 0-4.0). An applicant missing GRE AW still counts toward GPA, GRE and GRE V. Without the range filter the 'GRE' column averages 261 because most applicants typed their combined GRE total into it.

Question 4: Average GPA of American Fall 2026 applicants

Question. What is the average GPA of American applicants who applied for Fall 2026?

Result. Average GPA of American Fall 2026 applicants: 3.79

SQL.

```
SELECT ROUND(AVG(gpa)::numeric, 2)
FROM applicants
WHERE term ILIKE 'fall 2026'
      AND us_or_international ILIKE 'american'
      AND gpa > 0 AND gpa <= 4.0;
```

Explanation. Filters to Fall 2026 rows classified as American that report a GPA on the 4.0 scale (the range test also excludes NULLs).

Question 5: Fall 2025 acceptance percentage

Question. What percentage of Fall 2025 entries are acceptances?

Result. Fall 2025 acceptance percentage: 40.74%

SQL.

```
SELECT ROUND(100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%') / NULLIF(COUNT(*), 0), 2)
FROM applicants
WHERE term ILIKE 'fall 2025';
```

Explanation. Denominator: all Fall 2025 entries. Numerator: those whose cleaned status begins with 'Accept' (case-insensitive). Multiplying by 100.0 forces decimal arithmetic.

Question 6: Average GPA of accepted Fall 2026 applicants

Question. What is the average GPA of accepted applicants who applied for Fall 2026?

Result. Average GPA of accepted Fall 2026 applicants: 3.76

SQL.

```
SELECT ROUND(AVG(gpa)::numeric, 2)
FROM applicants
WHERE term ILIKE 'fall 2026'
      AND status ILIKE 'accept%'
      AND gpa > 0 AND gpa <= 4.0;
```

Explanation. Same shape as Question 4 but the second filter is on status instead of nationality.

Question 7: JHU Computer Science master's entries

Question. How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result. JHU Computer Science master's entries: 20

SQL.

```
SELECT COUNT(*)
FROM applicants
WHERE (program ILIKE '%johns hopkins%' OR program ILIKE '%john hopkins%' OR program ~* '\mJHU\M')
      AND program ILIKE '%computer science%'
      AND (degree ILIKE 'master%' OR degree ILIKE 'ms' OR degree ILIKE 'm.s.%');
```

Explanation. Uses the original program text. The university test accepts 'Johns Hopkins', the common misspelling 'John Hopkins', and the whole-word abbreviation 'JHU'; the program test looks for 'computer science' anywhere in the text; the degree test accepts Masters/MS variants.

Question 8: Fall 2026 CS PhD acceptances at four universities (original fields)

Question. How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford, or Carnegie Mellon?

Result. Fall 2026 CS PhD acceptances at four universities (original fields): 30

SQL.

```
SELECT COUNT(*)
FROM applicants
WHERE term ILIKE 'fall 2026'
      AND status ILIKE 'accept%'
      AND degree ILIKE 'phd'
      AND program ILIKE '%computer science%'
      AND (program ILIKE '%georgetown%' OR program ILIKE '%massachusetts institute of technology%' OR program ~* '\
```

Explanation. All five restrictions are ANDed together. The university clause matches the original program text with ILIKE for the spelled-out names and a whole-word regex for 'MIT'.

Question 9: Same as Q8 using the LLM-standardized fields

Question. Repeat Question 8 identifying the university and program with llm_generated_program / llm_generated_university.

Result. Same as Q8 using the LLM-standardized fields: 30

Original-field count: 30 | LLM-field count: 30 | Difference: +0

SQL.

```
SELECT COUNT(*)
FROM applicants
WHERE term ILIKE 'fall 2026'
      AND status ILIKE 'accept%'
      AND degree ILIKE 'phd'
      AND llm_generated_program ILIKE '%computer science%'
      AND (llm_generated_university ILIKE 'georgetown university' OR llm_generated_university ILIKE 'massachusetts i
```

Explanation. Term, status and degree still come from the original fields; program and university now use the LLM-standardized columns, so abbreviations, typos and campus variants that the free-text search in Q8 could miss (or

over-match) are resolved to canonical names first.

Why Q8 and Q9 agree here: Grad Cafe's current page layout already shows the university in its own column, so the Module 2 pipeline stored a clean university name for every row and the LLM/canonical post-processing kept it. For these four well-known universities the free-text search of the original 'program, university' string and the exact match on the standardized name therefore select exactly the same rows (verified by comparing the p_id sets). The counts would differ if applicants had typed abbreviations or misspellings that the original-field search misses (e.g. 'Carnegie Melon') or if the regex for 'MIT' over-matched an unrelated program - which is precisely what the standardized columns are meant to protect against.

Question 10: Acceptance rate by degree type (original question 1)

Question. Do PhD applicants report a different acceptance rate than master's applicants? For each degree type with at least 500 entries, what percentage of entries are acceptances?

Result.

Degree	Entries	Acceptance %
PhD	33,156	24.38%
Masters	14,680	67.59%
MFA	1,107	35.59%
PsyD	580	36.03%

SQL.

```
SELECT degree,
       COUNT(*) AS entries,
       ROUND(100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%') / COUNT(*), 2) AS accept_pct
FROM applicants
WHERE degree IS NOT NULL
GROUP BY degree
HAVING COUNT(*) >= 500
ORDER BY entries DESC;
```

Explanation. Groups all rows by degree, keeps only degree types with a meaningful sample (≥ 500 rows), and computes the share of accepted entries within each group with a FILTERed count.

Question 11: Most-reported universities for Fall 2026 (original question 2)

Question. Which ten universities have the most Fall 2026 entries, and what is the acceptance percentage and average reported GPA at each?

Result.

University	Entries	Acceptance %	Avg GPA
Stanford University	769	18.34%	3.85
University of California, Berkeley	728	25.14%	3.83
Yale University	633	14.38%	3.84
University of Washington	624	25.80%	3.77
University of Wisconsin–Madison	603	19.90%	3.81
Princeton University	600	20.50%	3.86
University of Michigan, Ann Arbor	584	25.68%	3.82
University of Chicago	580	27.24%	3.82
Massachusetts Institute of Technology	577	18.54%	3.85
University of Toronto	535	39.81%	3.79

SQL.

```
SELECT llm_generated_university AS university,
       COUNT(*) AS entries,
       ROUND(100.0 * COUNT(*) FILTER (WHERE status ILIKE 'accept%') / COUNT(*), 2) AS accept_pct,
```

```
        ROUND(AVG(gpa) FILTER (WHERE gpa > 0 AND gpa <= 4.0)::numeric, 2) AS avg_gpa
FROM applicants
WHERE term ILIKE 'fall 2026'
      AND llm_generated_university IS NOT NULL
      AND llm_generated_university <> 'Unknown'
GROUP BY llm_generated_university
ORDER BY entries DESC
LIMIT 10;
```

Explanation. Uses the LLM-standardized university so spelling variants collapse into one group, aggregates Fall 2026 rows per university, and ranks by volume. AVG(gpa) again uses only rows that reported a GPA.